Institute of Mathematical Sciences
University of Malaya

SIM1001 Basic Mathematics
SIX1013 Fundamentals of Advanced Mathematics

TUTORIAL 2 (Part 1)

1. Rewrite each statement in (i)-(v) in "if-then" form.
   (i) $2a - 5 = 11$ implies $a = 8$.
   (ii) 111111 is prime only if it is not a multiple of 4.
   (iii) Receipt will be not issued unless you clear your balance.
   (iv) Having two 45 angles is a sufficient condition for this triangle to be a right triangle.
   (v) Being divisible by 3 is a necessary condition for this number to be divisible by 9.

2. Write the negation, converse and contrapositive for each of the following statements. (Assume a is a particular real number)
   (i) If P is a square, then it is a rectangle.
   (ii) If $a \neq 1$ and $a \neq 0$ then, $a^2 \neq a$.
   (iii) If a is nonnegative, then a is positive or a is 0.
   (iv) If $a = 17$ or $a^3 = 8$, then a is prime.

3. Use truth table, show that $\sim(p \leftrightarrow q) \equiv p \leftrightarrow \sim q \equiv \sim p \leftrightarrow q$.

4. Use the laws of logical equivalences, show that $p \to (q \to r) \equiv (p \wedge q) \to r$.

5. Write the negation for each of the following statements.
   (i) No even number is prime. (Domain: The set of all integers.)
   (ii) At least one odd number that is a factor of 45 but not a factor of 21.
   (Domain: The set of all odd numbers.)
   (iii) $\forall x \in \mathbb{R}$, if $x > 3$, then $x^2 > 9$.
   (iv) $\exists x \in \mathbb{R}$ such that $x^2 - 2 \geq 0$.

6. Use truth table to determine the validity of each of the following arguments.

(i) If A is in team X, then B is in Team Y.
If C is in team X, then A is in Team X.
B is not in Team Y.
$\therefore$ C is not in Team X.

(ii) If x or y is divisible by 6, then xy is divisible by 6.
Neither of x nor y is divisible by 6.
$\therefore$ xy is not divisible by 6.

(iii) If A gets the new job, he will stay in Kuala Lumpur.
A will not buy a new car unless he stays in Kuala Lumpur.
A neither got the new job nor bought a new car.
$\therefore$ A did not stay in Kuala Lumpur.

7. Use some valid argument forms to deduce the conclusion from the premises for each of the following arguments.

(i)
$$\begin{array}{l} (p \vee q) \to \sim r \\ \sim p \wedge s \\ \sim s \vee r \\ \hline \therefore \sim q \end{array}$$

(ii)
$$\begin{array}{l} p \to q \vee r \\ q \to u \\ w \to \sim r \\ \sim u \wedge w \\ \hline \therefore\ \sim p \end{array}$$

(iii)
$$\begin{array}{l} \sim p \vee q \to r \\ s \vee \sim q \\ \sim u \\ p \to u \\ \sim p \wedge r \to \sim s \\ \hline \therefore \sim q \end{array}$$

8. Prove (i) – (iv), by mathematical induction:

(i) $\displaystyle\sum_{u=1}^{n} u^3 = \left(\frac{n(n+1)}{2}\right)^2, \quad \forall n \in \mathbb{Z}^+.$

(ii) $(1+t)^n > 1 + nt, \qquad \forall t > 0,\ n \geq 2,\ n \in \mathbb{Z}^+.$

(iii) $\displaystyle\sum_{r=1}^{n} r(r!) = (n+1)! - 1, \quad \forall n \in \mathbb{Z}^+.$

(iv) $\displaystyle(\cos\theta)(\cos 2\theta)(\cos 4\theta)\cdots(\cos 2^{n-1}\theta) = \frac{\sin 2^n\theta}{2^n \sin\theta}, \quad \forall n \in \mathbb{Z}^+.$

9. (i) Expand $(x+3y)^4$, $(2x-y)^5$ and $(2x+y^2)^6$.

(ii) Given that $(a+b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6ab^5 + b^6$, expand $(a+b)^7$ and $(a+b)^8$.

(iii) Without expanding $(2x^3 - y^2)^8$, find the term containing x^{12}.

(iv) Find the coefficient of $x^{16}y^4$ in the expansion of $\left(x - \dfrac{y}{2}\right)^{20}$.

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